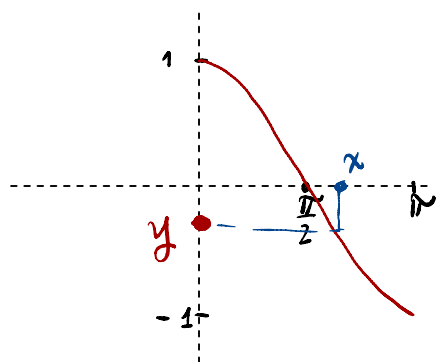


Lecture 7, Feb 04 - 2026.

The inverse cosine function:

Let $\cos x$ for $0 \leq x \leq \pi$.



in this case
 $\cos^{-1} x = y \stackrel{\text{def}}{\Leftrightarrow} \cos y = x, \quad 0 \leq y \leq \pi.$

$$\cos^{-1}(\cos(x)) = x \quad \text{for } 0 \leq x \leq \pi.$$

$$\cos(\cos^{-1}(y)) = y \quad \text{for } -1 \leq y \leq 1$$

$$\text{Dom}(\cos^{-1}) = [-1, 1], \quad \text{Range}(\cos^{-1}) = [0, \pi].$$

$$\frac{d}{dx}(\cos^{-1}(x)) = \frac{-1}{\sqrt{1-x^2}} \quad -1 < x < 1.$$

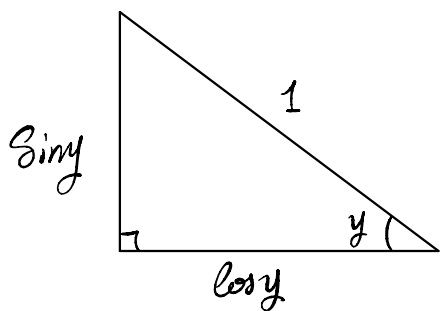
We know that $\cos(\cos^{-1}x) = x$

$$\frac{d}{dx}(\cos(\cos^{-1}x)) = \frac{d}{dx}(x) = 1.$$

$$-\sin(\cos^{-1}(x)) \cdot \frac{d}{dx}(\cos^{-1}x) = 1.$$

$$\frac{d}{dx}(\cos^{-1}x) = \frac{-1}{\sin(\cos^{-1}x)} = \frac{-1}{\sin y}.$$

Let $y = \cos^{-1}(x)$, $0 \leq y \leq \pi$.



$$\sin^2 y + \cos^2 y = 1.$$

$$\sin^2 y = 1 - \cos^2 y.$$

$$\sin y = \sqrt{1 - \cos^2 y} = \sqrt{1 - [\cos(\cos^{-1}x)]^2}$$

$$\sin y = \sqrt{1-x^2}$$

$$\frac{d}{dx} (\cos^{-1}(x)) = \frac{-1}{\sqrt{1-x^2}}$$

The inverse of the tangent function.

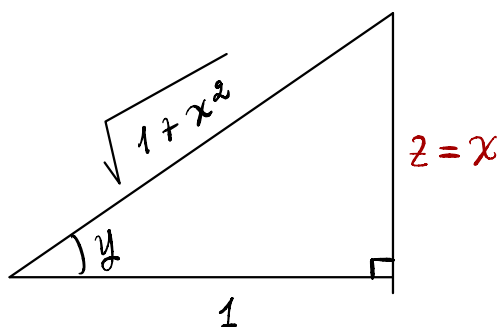
Let $f(x) = \tan x$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$. So,

$$\tan^{-1} x = y \quad (\Rightarrow) \quad \tan y = x \quad -\frac{\pi}{2} < y < \frac{\pi}{2}$$

Find: $\frac{d}{dx} (\tan^{-1} x)$. For you!

Example: Simplify the expression $\cos(\tan^{-1} x)$.

Sol: Let $y = \tan^{-1} x$ then $\tan y = x$. $-\frac{\pi}{2} < y < \frac{\pi}{2}$.



We know that -

$$\sec^2 y = 1 + \tan^2 y = 1 + x^2$$

$$\sec y = \frac{\sqrt{1+x^2}}{1}$$

By P.T.

$$(\sqrt{1+x^2})^2 = 1^2 + x^2$$

$$1+x^2 = 1+x^2$$

$$x^2 = x^2$$

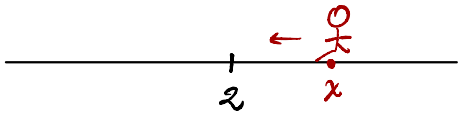
$$x = x$$

$$\text{then } \cos(y) = \cos(\tan^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

Prop: $\lim_{x \rightarrow -\infty} \tan^{-1} x = -\frac{\pi}{2}$ and $\lim_{x \rightarrow +\infty} \tan^{-1}(x) = \frac{\pi}{2}$.

$$\lim_{x \rightarrow \frac{\pi}{2}^-} \tan x = \infty \quad \text{and} \quad \lim_{x \rightarrow -\frac{\pi}{2}^+} \tan x = -\infty$$

Example: Evaluate $\lim_{x \rightarrow 2^+} \arctan\left(\frac{1}{x-2}\right)$.

Sln: Let $t = \frac{1}{x-2}$.  $x-2 > 0$

then $t \rightarrow \infty$ as $x \rightarrow 2^+$

$$\lim_{x \rightarrow 2^+} \arctan\left(\frac{1}{x-2}\right) = \lim_{t \rightarrow \infty} \arctan(t) = \frac{\pi}{2}.$$

Review: $y = \csc^{-1}(x)$, $|x| \geq 1 \Rightarrow \csc y = x$ $y \in (0, \frac{\pi}{2}] \cup (\pi, \frac{3\pi}{2}]$
 $y = \sec^{-1}(x)$, $|x| \geq 1 \Rightarrow \sec y = x$ $y \in [0, \frac{\pi}{2}) \cup (\pi, \frac{3\pi}{2})$
 $y = \cot^{-1}(x)$, $|x| \geq 1 \Rightarrow \cot y = x$ $y \in (0, \pi)$

Table of Derivatives of inverse Trigonometric Functions.

$$\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\csc^{-1}(x)) = \frac{-1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\cos^{-1}x) = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\sec^{-1}(x)) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\tan^{-1}x) = ?$$

$$\frac{d}{dx}(\cot^{-1}(x)) = \frac{-1}{1+x^2}$$

Example: Differentiate $y = \frac{1}{\sin^{-1}x}$.

Sln:

$$\begin{aligned} y' &= \frac{d}{dx}((\sin^{-1}x)^{-1}) = -1(\sin^{-1}x)^{-2} \frac{d}{dx}(\sin^{-1}x) \\ &= \frac{-1}{(\sin^{-1}x)^2} \cdot \frac{1}{\sqrt{1-x^2}} \end{aligned}$$

Example: Prove the identity $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$.
for me on Friday!

* Integrals involving inverse trigonometric functions.

Prop:

- $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}(x) + C.$

- for you: $\int \boxed{} dx = \tan^{-1}x + C.$

Example: Find $\int_0^{1/4} \sqrt{\frac{1}{1-4x^2}} dx.$

Soln:

$$\begin{aligned}\int_0^{1/4} \sqrt{\frac{1}{1-4x^2}} dx &= \int_0^{1/4} \frac{1}{\sqrt{1-4x^2}} dx \\&= \int_0^{1/4} \frac{1}{\sqrt{1-(2x)^2}} dx \quad \text{let } u=2x \\&\quad du=2dx \\&\quad dx=\frac{1}{2} du \\&= \int_0^{1/2} \frac{1}{\sqrt{1-u^2}} \cdot \frac{1}{2} du \\&= \frac{1}{2} \int_0^{1/2} \frac{1}{\sqrt{1-u^2}} du \\&= \frac{1}{2} \sin^{-1}(u) \Big|_0^{1/2} \\&= \frac{1}{2} \left[\sin^{-1}\left(\frac{1}{2}\right) - \sin^{-1}(0) \right] \\&= \frac{1}{2} \left[\frac{\pi}{6} - 0 \right] = \frac{\pi}{12}.\end{aligned}$$

